

Next-Generation Fraud Detection

Protecting Our Bottom Line

An AI-Driven Approach to Real-Time Transaction
Monitoring

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| THE BOTTOM LINE UP FRONT



The Problem

Legacy rules systems miss subtle fraud and drive away customers via false alarms.



Our Solution

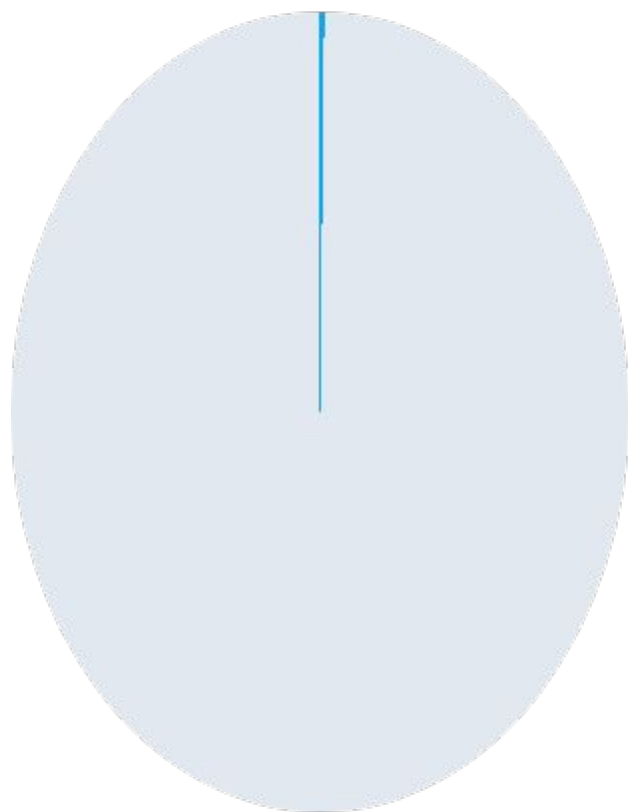
Custom ML engine analyzing behavioral patterns in milliseconds for instant precision.



The Impact

Maximizes fraud capture while actively reducing friction for legitimate customers.

BALANCING RISK AND REVENUE



The "Needle in a Haystack"

Fraud represents only **~0.17%** of our volume, making broad rules dangerous.

 **Actual Fraud (0.17%)**

 **Legitimate (99.83%)**

Declining a legitimate purchase frustrates top customers, causing "top-of-wallet" loss to competitors.

FROM STATIC RULES TO ADAPTIVE LEARNING

Flexible Intelligence

Instead of rigid "If/Then" logic, our AI trains on historical data to recognize sophisticated patterns.

Advanced Preprocessing

We use SMOTE (Synthetic Minority Over-sampling) to simulate rare fraud patterns, allowing the AI to learn how to block attacks before they occur.



PRECISION AT SCALE



Milliseconds

Analyze hundreds of anonymized data points simultaneously in real-time.



Probability

Outputs a granular score rather than a binary yes/no for business control.



Scalability

Elastic response during high-volume periods like holiday shopping seasons.

| MAXIMIZING PROTECTION & IMPACT

95

Target Capture

%

Reduced Chargebacks: Catching complex fraud charges that legacy rules completely miss.



Minimizing Friction: Approving more legitimate transactions without manual intervention.

| NO "BLACK BOX" MODELS



Full Auditability

Financial regulators require explanations for declined transactions. Our AI includes an "explainability layer."

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)]$$

Using SHAP values, we can instantly identify exactly which behavioral factors triggered an alert.

FAIR ALGORITHMS BUILD TRUST



Proactive Fairness Audit: We ensure models do not unintentionally "redline" or penalize specific customer demographics.



Zero Variance Goal: Internal tests proved a variance of less than 0.1% in error rates across age cohorts.



Brand Reputation: Proactive auditing protects our brand and shields us from heavy regulatory scrutiny.

EMPOWERING ANALYSTS

GenAI Copilot Integration

When the model blocks a transaction, the GenAI instantly writes a plain-English summary explaining why it looks suspicious.

This bridges the gap between complex ML and operational teams, drastically speeding up investigation times and training cycles.

Request URL

```
http://127.0.0.1:8000/predict
```

Server response

Code

Details

200

Response body

```
{  
  "is_fraud": true,  
  "fraud_probability": 0.985,  
  "status": "Transaction Blocked"  
}
```

Response headers

```
content-length: 74  
content-type: application/json  
date: Tue, 17 Mar 2026 07:06:41 GMT  
server: uvicorn
```

| EXECUTION ROADMAP



Questions?

Transforming Risk into Competitive Advantage

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Next-Gen Fraud Detection Initiative